Home / Grade 8 / Geometry / Quarter 1

Rectangle and its properties

A parallelogram with one right angle — which turns out to force all four, and makes the diagonals equal.

LESSON 7 | 1 × 40 MIN | GEOMETRY 8, TEMA 4 | STAGE 9 · BEYOND

LESSON CLOCK

Warm-up	4 min
Explanation	12 min
Interactive	8 min
Practice	14 min
Homework	2 min

LEARNING OBJECTIVES

- Define a rectangle and show that one right angle forces four.
- Prove that the diagonals of a rectangle are equal.
- Use the fact that the diagonals bisect each other and are equal.
- Apply Pythagoras to find a diagonal from the sides.

TEXTBOOK

National	Тема 4, pp. 14–15
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

Definition

DEFINITION

A **rectangle** is a parallelogram with a right angle.

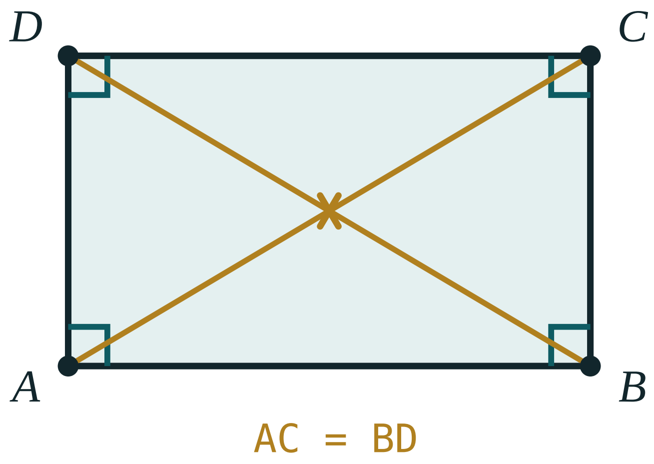
One right angle is enough. Neighbouring angles of a parallelogram are supplementary, so if $\angle A = 90°$ then $\angle B = 90°$, and opposite angles give $\angle C = \angle D = 90°$.

Because a rectangle *is* a parallelogram, it inherits everything from Тема 2: opposite sides equal, opposite angles equal, diagonals bisecting each other. Only the new property has to be proved.

The new property — equal diagonals

THEOREM

The diagonals of a rectangle are equal: $AC = BD$.



Four right angles, and two diagonals of exactly the same length crossing at the centre.

Proof. Compare $\triangle ABC$ and $\triangle BAD$:

- AB is common;
- $BC = AD$ — opposite sides of a parallelogram;
- $\angle ABC = \angle BAD = 90°$.

So $\triangle ABC \cong \triangle BAD$ by SAS, and therefore $AC = BD$. ∎

CONSEQUENCE

The diagonals bisect each other **and** are equal, so all four half-diagonals are equal: $AO = BO = CO = DO$. The point O is the same distance from all four vertices — it is the centre of the circle through them.

The test

TEST

A parallelogram whose diagonals are equal is a rectangle.

Note carefully: *a parallelogram* whose diagonals are equal. Without that word the statement is false — an isosceles trapezium also has equal diagonals.

The diagonal from the sides

A diagonal cuts the rectangle into two right-angled triangles, so Pythagoras applies:

$$d = \sqrt{a^2 + b^2}$$

A 6×8 rectangle has diagonal $\sqrt{36 + 64} = \sqrt{100} = 10$.

And the other two familiar formulae: perimeter $P = 2(a + b)$, area $S = ab$.

02 Worked examples

EXAMPLE 1 **A rectangle has sides 9 cm and 12 cm. Find its diagonal, perimeter and area.**

1. $d = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225}$

   Pythagoras on half the rectangle.

2. $d = 15\ cm$

3. $P = 2(9 + 12) = 42\ cm$

4. $S = 9 \cdot 12 = 108\ cm^2$

ANSWER

$d = 15\ cm,\ P = 42\ cm,\ S = 108\ cm^2$

EXAMPLE 2 ***In rectangle ABCD the diagonals meet at O and ∠AOB = 60°. Find ∠ACB.***

1. $AO = BO$

   Half-diagonals of a rectangle are all equal.

2. $\triangle AOB$ is isosceles with apex $60°$, so it is equilateral.

   All three angles are $60°$.

3. $\angle OAB = 60°$

   That is $\angle CAB$.

4. $\angle ACB = 90° - 60° = 30°$

   The acute angles of right triangle ABC add to $90°$.

ANSWER

$\angle ACB = 30°$

EXAMPLE 3 **The diagonal of a rectangle is 13 cm and one side is 5 cm. Find the area.**

1. $b = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144}$

   Pythagoras, rearranged.

2. $b = 12\ cm$

3. $S = 5 \cdot 12 = 60\ cm^2$

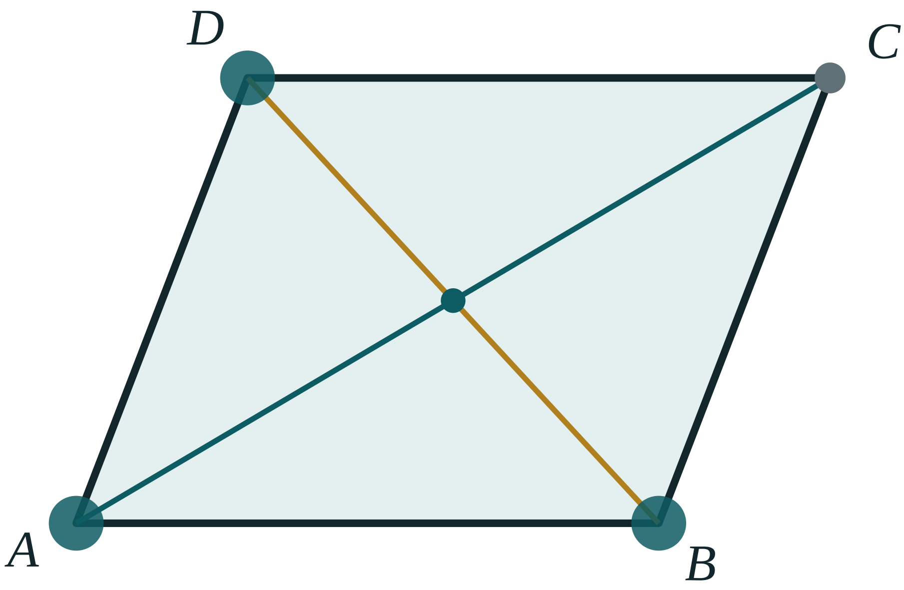
ANSWER

$60\ cm^2$

03 Interactive model

Drag until ∠A reads 90° and watch AC and BD become equal — then drag away and watch them separate again.

When do the diagonals become equal?



∠A **69°** | AC **255.5** | BD **176.9** | AO **127.8** | OC **127.8** | AB **170** | AD **139.3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rectangle	To‘g‘ri to‘rtburchak	Прямоугольник
Right angle	To‘g‘ri burchak	Прямой угол
Equal diagonals	Teng diagonallar	Равные диагонали
Area	Yuza	Площадь
Pythagoras’ theorem	Pifagor teoremasi	Теорема Пифагора
Hypotenuse	Gipotenuza	Гипотенуза
Circumscribed circle	Tashqi chizilgan aylana	Описанная окружность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A rectangle is defined as:

- a quadrilateral with a right angle
- a parallelogram with a right angle
- a parallelogram with equal sides
- a quadrilateral with equal diagonals

In a rectangle the diagonals are:

- perpendicular
- equal and bisect each other
- equal but do not bisect
- angle bisectors

A rectangle has sides 8 and 15. Its diagonal is:

- 17
- 23
- $\sqrt{23}$
- 120

A quadrilateral with equal diagonals must be a rectangle:

- true
- false — it could be an isosceles trapezium
- true only if it is convex
- true only if the sides are equal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *A rectangle has sides 3 cm and 4 cm. Find its diagonal.*

   ANSWER *5 cm*

2. *Sides 6 cm and 8 cm. Find the diagonal.*

   ANSWER *10 cm*

3. *Sides 5 cm and 7 cm. Find the perimeter.*

   ANSWER *24 cm*

4. *Sides 5 cm and 7 cm. Find the area.*

   ANSWER *35 cm²*

5. *AC = 10 cm in rectangle ABCD. Find BD.*

   ANSWER *10 cm*

6. *AC = 10 cm and the diagonals meet at O. Find BO.*

   ANSWER *5 cm*

7. *One angle of a rectangle is 90°. Find the other three.*

   ANSWER *90°* each

Medium · the standard question

7 PROBLEMS

1. *Sides 9 cm and 12 cm. Find the diagonal, perimeter and area.*

   ANSWER *15 cm, 42 cm, 108 cm²*

2. *The diagonal is 13 cm and one side is 5 cm. Find the area.*

   ANSWER *60 cm²*

3. *The perimeter is 34 cm and one side is 5 cm. Find the diagonal.*

   ANSWER other side *12 cm*, diagonal *13 cm*

4. *The diagonals meet at O and ∠AOB = 60°. Find ∠ACB.*

   ANSWER *30°*

5. *The area is 48 cm² and one side is 6 cm. Find the perimeter.*

   ANSWER *28 cm*

6. *The diagonals of a rectangle are 20 cm. Find the distance from O to each vertex.*

   ANSWER *10 cm*

7. *A rectangle has a diagonal of 25 cm and a side of 20 cm. Find the perimeter.*

   ANSWER other side *15 cm*, *P = 70 cm*

Hard · stretch and reason

7 PROBLEMS

1. *The diagonals meet at O and $\angle AOD = 120°$. Find the acute angle between a diagonal and the longer side.*

   ANSWER $\angle AOB = 60°$, so $\triangle AOB$ is equilateral and the angle is $60°$.

2. *A rectangle has perimeter 46 cm and diagonal 17 cm. Find its sides.*

   ANSWER $a + b = 23$, $a^2 + b^2 = 289 \Longrightarrow ab = 120 \Longrightarrow$ sides $8\ cm$ and $15\ cm$

3. *Prove that a parallelogram with equal diagonals is a rectangle.*

   ANSWER The diagonals bisect each other, so $\triangle ABC \cong \triangle BAD$ by SSS; then $\angle A = \angle B$ and they are supplementary, so each is $90°$.

4. *In rectangle ABCD the bisector of $\angle A$ meets BC at K, with BK = 4 cm and KC = 6 cm. Find the perimeter.*

   ANSWER $\triangle ABK$ is an isosceles right triangle, so $AB = BK = 4$ and $BC = 10$: $P = 28\ cm$.

5. *A rectangle is inscribed in a circle of radius 6.5 cm and one side is 5 cm. Find the other side.*

   ANSWER The diagonal is the diameter, $13\ cm$; the other side is $12\ cm$.

6. *Show that the point where the diagonals of a rectangle meet is equidistant from all four vertices.*

   ANSWER The diagonals bisect each other and are equal, so $AO = BO = CO = DO = \frac{d}{2}$.

7. *A rectangle has area 120 cm² and diagonal 17 cm. Find its sides.*

   ANSWER $ab = 120$ and $a^2 + b^2 = 289$, so $(a + b)^2 = 529$ and $a + b = 23$: the sides are $8\ cm$ and $15\ cm$.

07 Homework

Homework — 5 problems

Geometry 8, Тема 4, pp. 14–15.

1. *A rectangle has sides 7 cm and 24 cm. Find the diagonal, perimeter and area.*

2. *The diagonal is 20 cm and one side is 16 cm. Find the other side and the area.*
3. *The diagonals of rectangle ABCD meet at O and $\angle BOC = 120°$. Find $\angle OBC$.*
4. *The perimeter of a rectangle is 40 cm and one side is 4 cm longer than the other. Find the diagonal.*
5. *Prove that the diagonals of a rectangle are equal, stating the congruence test you use.*